



Systems in Animals & Plants

LEARNING COMPETENCIES

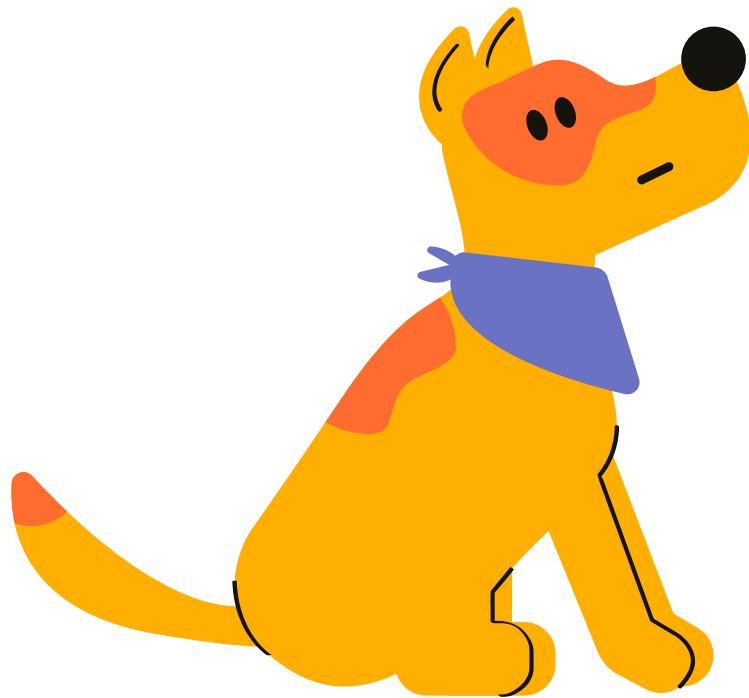
The learners can:

- Describe the skeletal, muscular, digestive, respiratory, and circulatory systems of animals and explain their roles in supporting movement, digestion, breathing, and transport of materials.
- Describe the root and shoot systems of plants and explain their roles in supporting plant growth and survival.

LET'S EXPLORE!

Questions to Wonder About

Q1



How does a dog run?

Q2



How does a bird breathe?

Q3

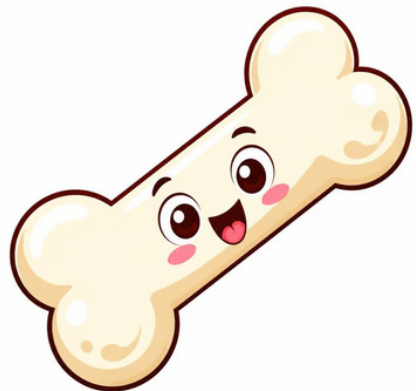


**How does a plant drink
water?**

PART 1

Animals Have Body Systems

Animals have body systems that work together to help them live and move.



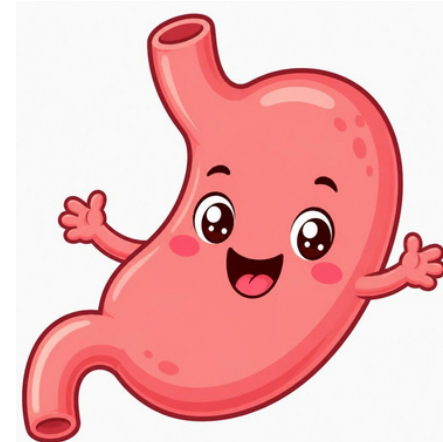
Skeletal

Bones



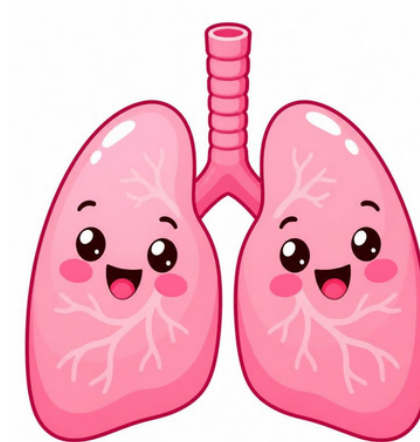
Muscular

Muscles



Digestive

Stomach



Respiratory

Lungs



Circulatory

Heart

SKELETAL SYSTEM

Bones Give Us Shape

The skeletal system supports and protects the body. Bones give the body shape and hold organs safely inside.

EXAMPLE

A dog's bones support its body and protect important organs like its heart and lungs.



MUSCULAR SYSTEM

Muscles Make Us Move

Muscles help animals move.
They work together with bones
so animals can walk, run, jump,
and play.

EXAMPLE

A dog uses strong muscles in its legs to sprint across the yard.



DIGESTIVE SYSTEM

Food Becomes Fuel

The digestive system breaks down food so the body can use the nutrients to grow strong and stay healthy.

EXAMPLE

When a dog eats, its digestive system works to turn the food into energy and nutrients.



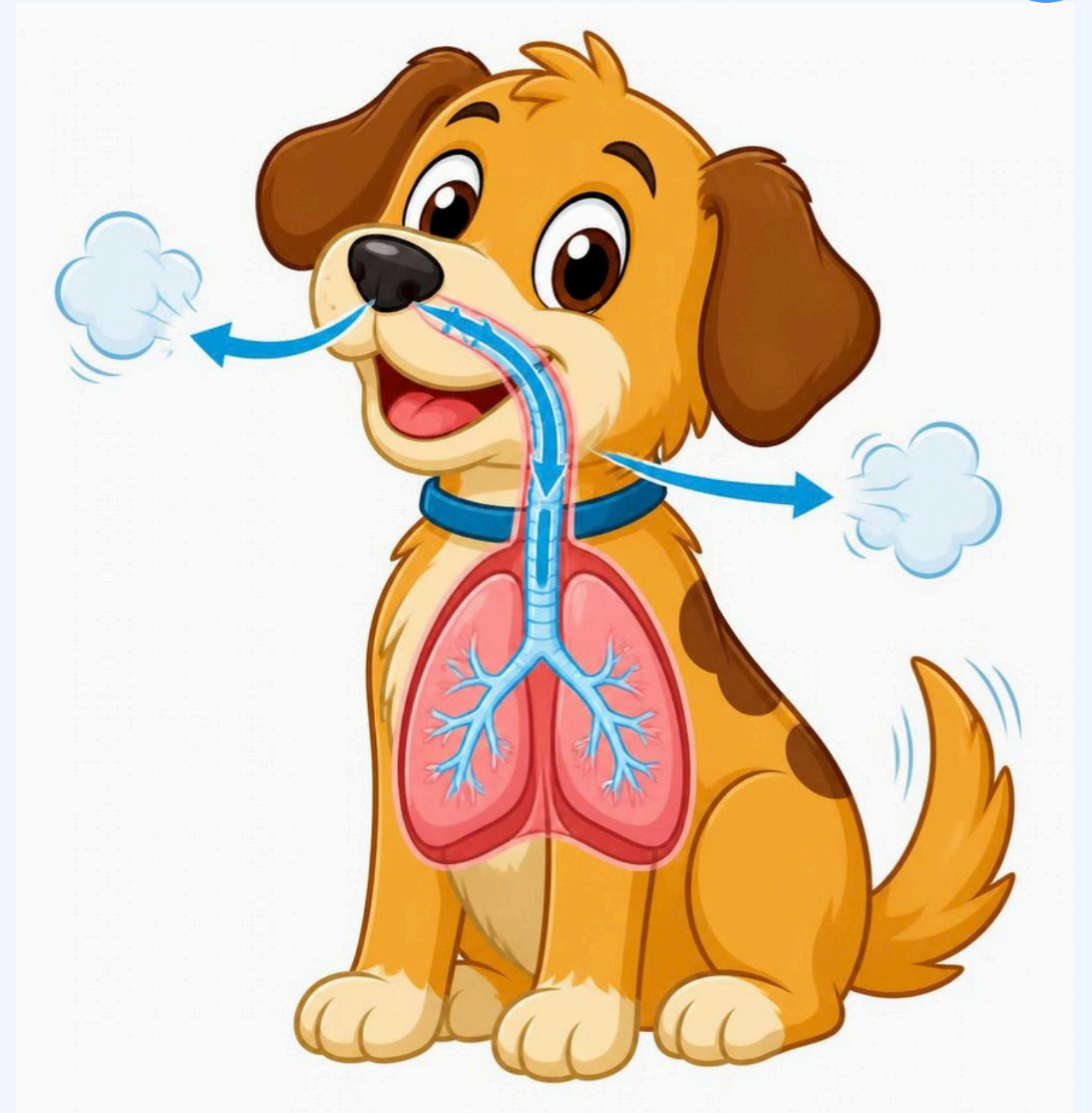
RESPIRATORY SYSTEM

Breathing In & Out

The respiratory system helps animals breathe. Lungs take in oxygen and remove carbon dioxide.

EXAMPLE

A dog breathes air in through its nose and mouth, filling its lungs with oxygen.



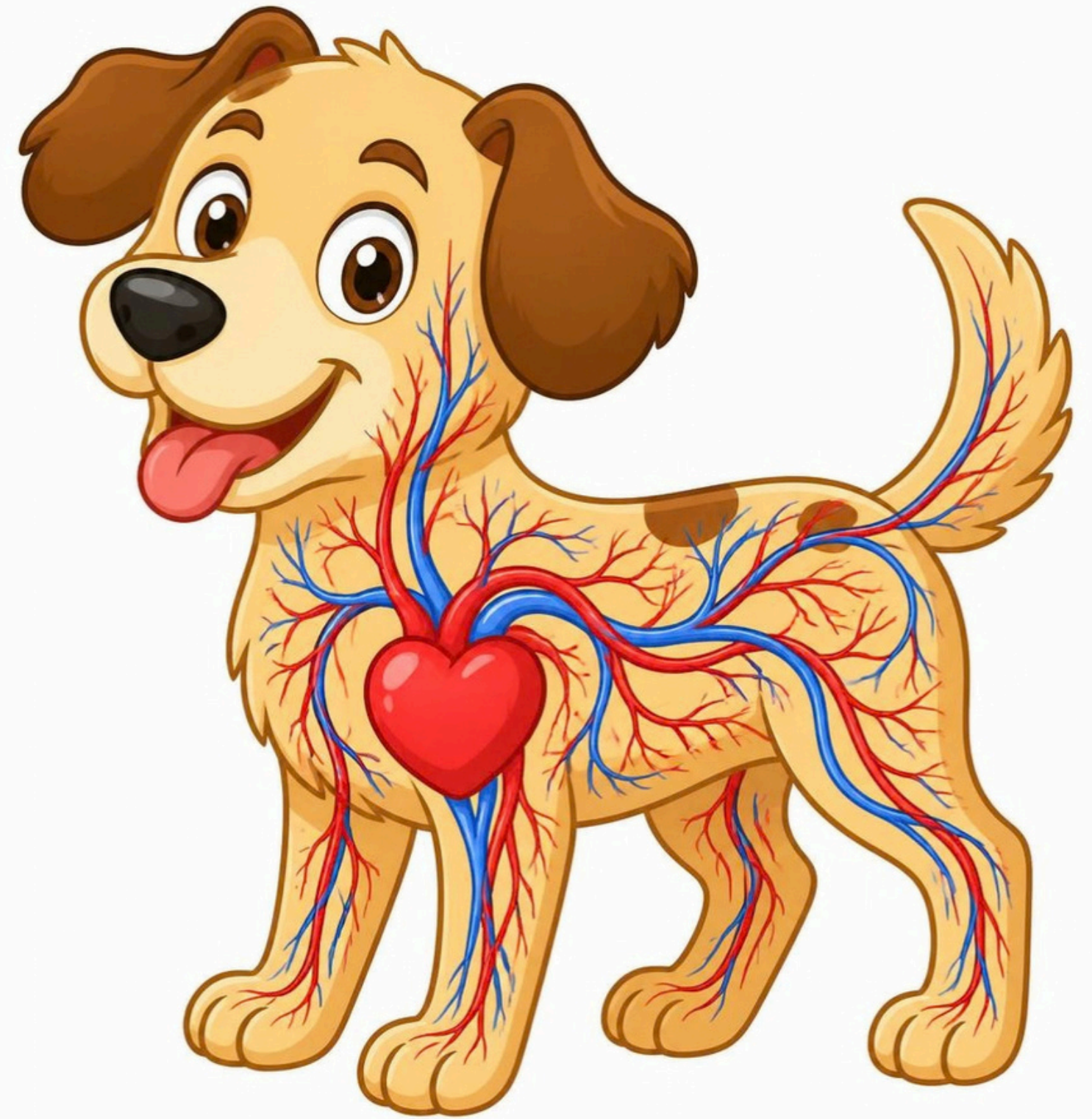
CIRCULATORY SYSTEM

The Body's Delivery Route

The circulatory system moves oxygen and nutrients through the body using the heart and blood vessels.

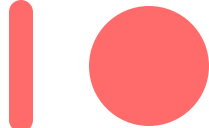
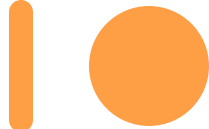
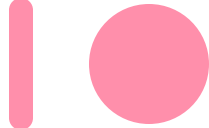
EXAMPLE

A dog's heart pumps blood through vessels all over its body, day and night.



RECAP

Animal Body Systems At A Glance

Body System	Main Function
 Skeletal	Supports the body and protects organs.
 Muscular	Works with bones to help animals move.
 Digestive	Breaks down food into usable nutrients.
 Respiratory	Takes in oxygen, removes carbon dioxide.
 Circulatory	Moves oxygen and nutrients around the body.



PART 2

Systems in Plants

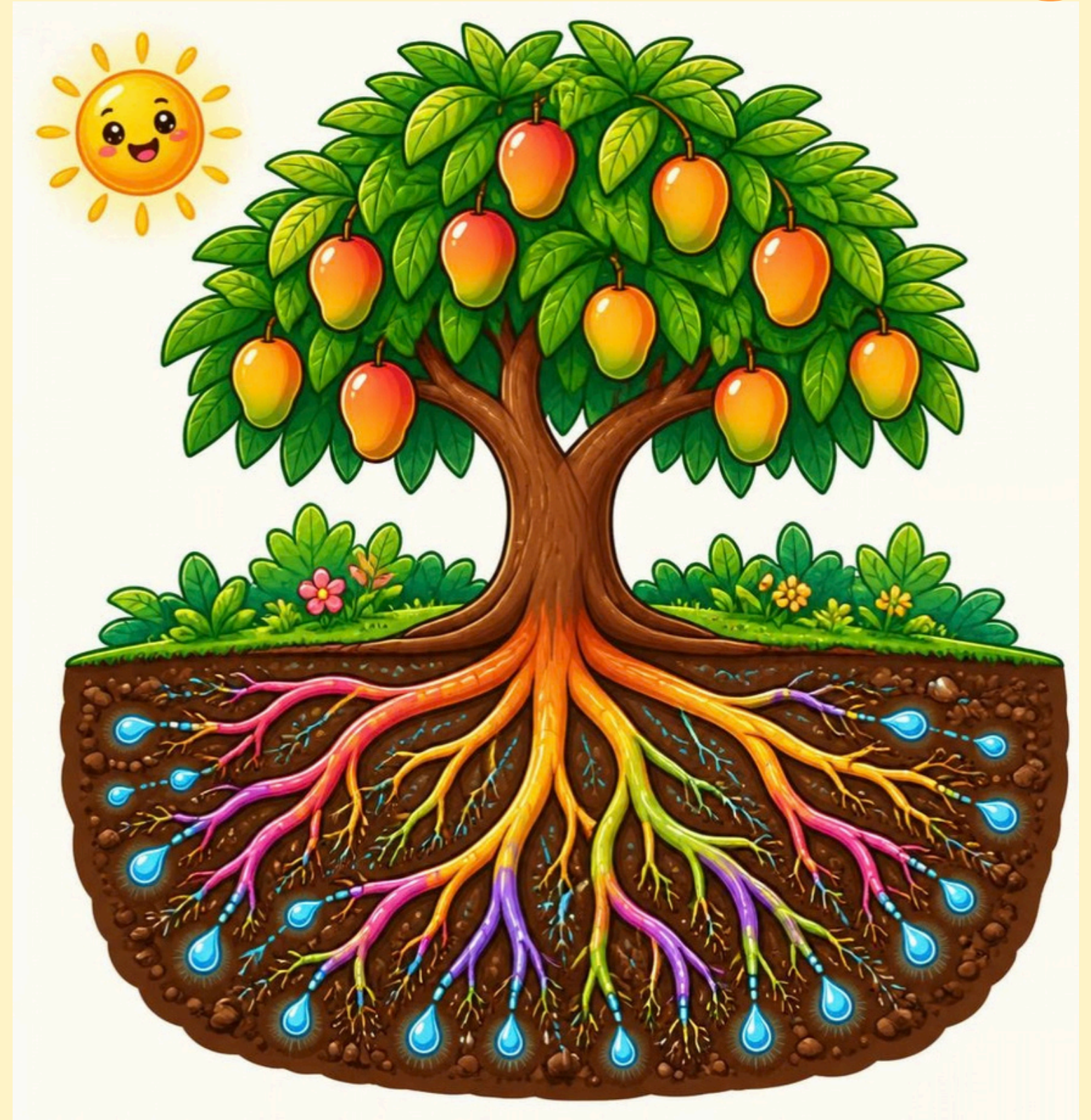
ROOT SYSTEM

Roots Anchor & Absorb

Roots hold the plant in the soil and soak up water and nutrients from the ground.

EXAMPLE

A mango tree's roots keep it steady in strong winds and gather water from deep in the soil.



SHOOT SYSTEM

Stems, Leaves & Flowers

The shoot system includes the stem, leaves, flowers, and fruits. It supports the plant and helps it grow toward the sun.

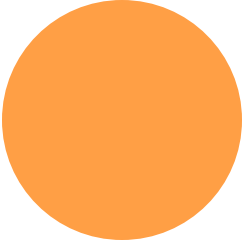
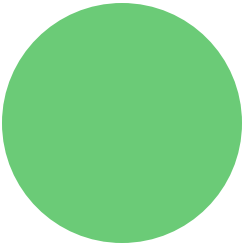
EXAMPLE

Leaves gather sunlight, flowers attract bees, and stems carry water up from the roots.



RECAP

Plant Systems At A Glance

Plant System		Main Roles
	Root System Roots	Anchors the plant and absorbs water and nutrients from the soil.
	Shoot System Stem, leaves, flowers, fruits	Supports the plant and carries out life processes like making food.

Animals & Plants: A System Showdown

Different systems, but both help living things survive and grow.

ANIMALS

-  **Skeletal**
-  **Muscular**
-  **Digestive**
-  **Respiratory**
-  **Circulatory**

PLANTS

-  **Root System**
-  **Shoot System**

INTERACTIVE ACTIVITY

Which System Is At Work?

Read each situation and guess which system is doing the job!

1 A dog uses its bones and muscles to run.

✓ **Skeletal + Muscular System**

2 A cat chews and swallows its food.

✓ **Digestive System**

3 A bird breathes in fresh air on a branch.

✓ **Respiratory System**

4 A rabbit's heart pumps blood after hopping.

✓ **Circulatory System**

5 A mango tree drinks water from the soil.

✓ **Root System**

6 A sunflower's leaves gather sunlight.

✓ **Shoot System**

ASSESSMENT

Quick Check: 5 Questions

Q1 Which animal system supports the body and protects organs?

A Muscular

B Skeletal

C Digestive

D Respiratory

Q2 Which system helps a dog breathe?

A Circulatory

B Digestive

C Respiratory

D Skeletal

Q3 Which system moves blood around the body?

A Circulatory

B Muscular

C Root

D Shoot

Q4 Which plant system absorbs water from the soil?

A Shoot

B Leaf

C Root

D Flower

Q5

ASSESSMENT

Quick Check: 5 Questions

Q5

Which is NOT part of the shoot system?

A

Stem

B

Leaves

C

Roots

D

Flowers

THANK
YOU!

